
Fundamental of Machine learning

REQUEST AND BEAUTIFULSOUP



Request

The *requests* module allows you to send HTTP requests using Python.

The HTTP request returns a Response Object with all the response data (content, encoding, status, etc).

```
import requests  
  
x = requests.get('https://google.com')  
  
print(x.text)
```

Request - Get

The `get()` method sends a GET request to the specified url.

```
requests.get(url, params={key: value}, args)
```

```
import requests
```

```
x = requests.get('https://google.com')
```

```
print(x.text)
```

Request – Get with Parameter

The `get()` method sends a GET request to the specified url.

```
requests.get(url, params={key: value}, args)
```

```
import requests
```

```
params = { 'src': 'tree' }
```

```
x = requests.get("https://tiki.vn/sach-tieng-anh/c320", params)
```

Request – Get with headers

The `get()` method sends a GET request to the specified url.

`requests.get(url, headers={key: value}, args)`

```
import requests

url = 'https://web.com'

#use the 'headers' parameter to set the HTTP headers:
x = requests.get(url, headers = {"HTTP_HOST": "MyVeryOwnHost"})

print(x.text)
```

Request – Post

The `post()` method sends a POST request to the specified url.

The `post()` method is used when you want to send some data to the server.

Request – Post with data

Optional. A dictionary, list of tuples, bytes or a file object to send to the specified url.

```
import requests

url = 'https://web.com'
myobj = {'somekey': 'somevalue'}

x = requests.post(url, data = myobj)

print(x.text)
```

Request – Post with Json

Optional. A JSON object to send to the specified url

```
import requests

url = 'https://www.w3schools.com/python/demopage.php'
myjson = {'somekey': 'somevalue'}

x = requests.post(url, json = myjson)

#print the response text (the content of the requested file):

print(x.status_code)
```


Request – Post with Files

Optional. A dictionary of files to send to the specified url.

```
import requests

url = 'https://web.com'
myobj = {'somekey': 'somevalue'}

x = requests.post(url, data = myobj)

print(x.text)
```

Beautiful Soup

Beautiful Soup is a Python library for pulling data out of HTML and XML files

Searching, and modifying the parse HTML or XML tree.

Install:

```
pip install beautifulsoup4
```

Beautiful Soup

Beautiful Soup apply html as a string

Example:

```
from bs4 import BeautifulSoup

html = '''
<html>
  <body>Hello world</body>
</html>
'''

soup = BeautifulSoup(html)
print(soup.prettify())
```

Beautiful Soup

Beautiful Soup working with file

Example:

```
from bs4 import BeautifulSoup
# Pass in an open filehandle
with open("index.html") as fp:
    soup = BeautifulSoup(fp)
    print(soup.prettify())
```

Beautiful Soup with Requests

Beautiful Soup is a Python library for pulling data out of HTML and XML files

```
from bs4 import BeautifulSoup
import requests
# Pass in a string
response = requests.get('http://data.pr4e.org/romeo.txt')
soup = BeautifulSoup(response.text, 'html.parser')
print(soup.prettify())
```

Beautiful Soup

Beautiful Soup transforms a complex HTML document into a complex tree of Python objects.

Kinds of objects

- Tag
- Name
- Attributes

Beautiful Soup

Kinds of objects

- Tag

```
from bs4 import BeautifulSoup

html = '''
<html>
  <body>Hello world</body>
</html>
'''

soup = BeautifulSoup(html)
# Get element by tag
tag = soup.body
print(tag)
```

Beautiful Soup

Kinds of objects

- Attributes

```
from bs4 import BeautifulSoup

html = '''
<html>
  <body id='body_id' class='content'>Hello world</body>
</html>
'''

soup = BeautifulSoup(html)
# Get element by tag
tag = soup.body
print(tag.attrs)
```


Beautiful Soup

You can add, remove, and modify a tag's attributes

```
from bs4 import BeautifulSoup

html = '''
<html>
  <body id='body_id' class='content'>Hello world</body>
</html>
'''
soup = BeautifulSoup(html)
# Get element by tag
tag = soup.body
tag.name = 'body_fa'
del tag['id']
tag['iidd'] = 'fa'
tag['class'] = 'fa'
print(soup.prettify())
```

Beautiful Soup

Navigating using tag names

```
from bs4 import BeautifulSoup

html = '''
<html>
  <body id='body_id' class='content'>
    <p>Hello world</p>
  </body>
</html>
'''

soup = BeautifulSoup(html)
# Get element by tag
body = soup.body
p_text = body.p.text
print(p_text)
```

Beautiful Soup

Searching the tree – *find_all(name, attrs, recursive, string, limit)*

```
from bs4 import BeautifulSoup
html = '''
<html>
  <body id='body_id' class='content'>
    <p>Hello world</p>
    <p> FA </p>
  </body>
</html>
'''
soup = BeautifulSoup(html)
# Get element by tag
list_p_tags = soup.find_all('p')
print(list_p_tags)
```

Beautiful Soup

Searching the tree – *find_all(name, attrs, recursive, string, limit)*

The *name* argument

```
from bs4 import BeautifulSoup
html = '''
<html>
  <body id='body_id' class='content'>
    <p>Hello world</p>
    <p> FA </p>
  </body>
</html>
'''
soup = BeautifulSoup(html)
# Get element by tag
list_p_tags = soup.find_all('p')
print(list_p_tags)
```

Beautiful Soup

Searching the tree – *find_all(name, attrs, recursive, string, limit)*

The *attribute* argument

```
from bs4 import BeautifulSoup

html = '''
<html>
  <body id='body_id' class='content'>
    <p class = 'p1'>Hello world</p>
    <p> FA </p>
  </body>
</html>
'''

soup = BeautifulSoup(html)
# Get element by tag
list_p_tags = soup.find_all('p', attrs= {'class': 'p1'})
print(list_p_tags)
```

Beautiful Soup

Searching the tree – *find_all(name, attrs, recursive, string, limit)*

The *recursive* argument

```
from bs4 import BeautifulSoup
html = '''
<html>
  <body id='body_id' class='content'>
    <p>Hello world</p>
    <p> FA </p>
  </body>
</html>
'''
soup = BeautifulSoup(html)
# Get element by tag
list_p_tags = soup.find_all('p', recursive=False)
print(list_p_tags)
```

Beautiful Soup

Searching the tree – *find_all(name, attrs, recursive, string, limit)*

The *string* argument

```
from bs4 import BeautifulSoup
import re
html = '''
<html>
  <body id='body_id' class='content'>
    <p class = 'p1'>Hello world</p>
    <p>Hello FA </p>
  </body>
</html>
'''
soup = BeautifulSoup(html)
list_values = soup.find_all(string=re.compile("Hello"))
print(list_values)
```

Beautiful Soup

Searching the tree – *find_all(name, attrs, recursive, string, limit)*

The *limit* argument

```
from bs4 import BeautifulSoup
import re
html = '''
<html>
  <body id='body_id' class='content'>
    <p class='p1'>Hello world</p>
    <p>Hello FA </p>
  </body>
</html>
'''
soup = BeautifulSoup(html)
list_values = soup.find_all(string=re.compile("Hello"), limit=1)
print(list_values)
```


Beautiful Soup

.contents and *.children*

```
from bs4 import BeautifulSoup
html = '''
<html>
  <body id='body_id' class='content'>
    <p class='p1'>Hello world</p>
    <p>Hello FA </p>
  </body>
</html>
'''
soup = BeautifulSoup(html)
list_values = soup.body.contents
for content in list_values:
    print(content)
```

Beautiful Soup

.contents and *.children*

```
from bs4 import BeautifulSoup
html = '''
<html>
  <body id = 'body_id' class = 'content'>
    <div>
      <p>FA </p>
    </div>
    <p class = 'p1'>Hello world</p>
    <p>Hello FA </p>
  </body>
</html>
'''
soup = BeautifulSoup(html)
list_values = soup.body.children
for content in list_values:
    print(content)
```

THANK YOU!
